

Learning Descriptive Statistics with Hantavirus Data

A Step-by-Step Journey Through Distribution Analysis

1 Introduction

This document follows my process of applying descriptive statistics concepts learned from a textbook to a real-world dataset – as a practice project to learn and improve. Along the way I share findings, mistakes made, and how they were corrected.

1.1 Data Source

Data is from Kaggle: <https://www.kaggle.com/datasets/zkskhurram/hantavirus-andes-virus-global-epidemiology>

Disclaimer: This is a historical Kaggle dataset on hantavirus epidemiology, **not** the 2025 outbreak data. This project focuses on learning descriptive statistics, not on current events.

Six datasets covering:

- **Clinical:** 7538 patient records (incubation, hospital stay, ICU, severity, outcome)
- **Environmental:** 896 location-quarters (temperature, rainfall, rodent abundance, biome)
- **Country-Yearly:** 939 country-year observations (confirmed cases, deaths, CFR)
- Outbreaks, virus strains, monthly trends

The data was already well-prepared. No further cleaning was needed beyond noting that missing symptom data was structural – each syndrome (HPS vs HFRS) uses a different symptom checklist, so no imputation was required.

1.2 Project Structure

1.3 Project Structure

hanta_project/	
scripts/	Python scripts (O1_, D1_, D2_, ..., E1_, F_)
figures/	All output plots (PDF + PNG)
explanations/	Per-step documentation
tables/	LaTeX summary tables
report/	This report

All scripts use `np.random.seed(42)` for reproducibility. Key dependencies: pandas, numpy, scipy, matplotlib, seaborn.

1.4 Concepts Applied

Box plots, violin plots, histograms, ECDF, Q-Q plots, subgroup analysis, outlier detection (IQR), mean vs median, skewness, kurtosis, Shapiro-Wilk test, variability measures (std, MAD, IQR, range).

2 Phase A: Setup

`scripts/helper.py` was created with shared functions: loading CSVs, IQR outlier detection, summary statistics printing, normality testing, and figure saving.

```
def iqr_outliers(series):  
    Q1, Q3 = series.quantile(0.25), series.quantile(0.75)  
    IQR = Q3 - Q1  
    return series[(series < Q1 - 1.5*IQR) | (series > Q3 + 1.5*IQR)]
```

3 Phase B: Data Exploration

3.1 B1 – Load and Inspect (`01_load_explore.py`)

Every CSV was loaded and checked: shape, dtypes, missing values, duplicates, categorical value counts, numeric ranges.

Key finding: Clinical symptom columns have two clear missing-data patterns:

- 65.2% missing: myalgia, cough, dyspnea, etc. — all HFRS patients
- 34.8% missing: back_pain, abdominal_pain, etc. — all HPS patients

This is structural (MAR), not random. Each syndrome uses a different symptom checklist. No imputation needed.

3.2 B2 – Summary Statistics (`02_summary_stats.py`)

For every numeric column: count, mean, median, std, variance, MAD, IQR, range, skewness, kurtosis.

Key findings from mean–median gaps:

- `confirmed_cases`: mean=1793, median=58 → extreme right skew
- `icu_days`: median=0, mean=3.3 → spike-and-slab
- `avg_temp_c`: mean=11.3, median=11.7 → near symmetric

Lesson: The gap between mean and median is the quickest skewness test.

4 Phase C: Outlier Investigation (C_outlier_investigation.py)

Applied IQR rule to every numeric column. Each flagged value was printed with context (country, severity, biome) to decide keep vs remove.

Verdict: All kept. No data errors found. Every flagged value is a genuine extreme explained by context (Critical patients, Valdivian rainfall, China HFRS).

IQR rule note: The IQR rule assumes roughly symmetric spread. On extreme right-skewed variables like confirmed_cases, $Q3 + 1.5 \times IQR$ falls within the bulk of the real tail rather than beyond it. For skewed data, log-transform then IQR, or MAD-based detection, can be more appropriate.

Note: When 18% of data is flagged (confirmed_cases), it's not outliers – it's a different subgroup (China HFRS). This is the same pattern as hospital_days by severity: always check subgroups.

5 Phase D: Univariate Distribution Analysis

Each variable was analyzed with the same workflow:

1. Box plot (median, IQR, outliers)
2. Violin plot with box overlay (distribution shape + summary)
3. Histogram (integer-aligned for discrete, FD rule for continuous)
4. ECDF with theoretical normal overlay
5. Q-Q plot (normality assessment; used only for unbounded variables)
6. Subgroup investigation
7. Shapiro-Wilk test (with $n > 5000$ caveat)
8. Questionnaire: user observes, then compares with numbers

5.1 D1: Incubation Days (`D1_incubation_days.py`)

Mistakes Made (documented in `D1_mistakes_and_fixes.md`)

Mistake 1: Beeswarm plot

- **What we used:** Strip plot with jitter but no vertical spread
- **Why it failed:** All points on one horizontal line — couldn't see density
- **Fix:** Replaced with box plot + violin plot with box overlay

Mistake 2: Too many histogram bins

- **What we used:** $\sqrt{7538} \approx 86$ bins for range 7–42 (35 days)
- **Why it failed:** Integer data at 0.4-day bins creates empty bins — "comb" effect
- **Fix:** Integer-aligned bins (edges at half-integers) for discrete data

Mistake 3: KDE on discrete data

- **What we used:** Seaborn KDE with bandwidth comparison
- **Why it failed:** KDE assumes continuous data. Density below 0 (impossible). Bandwidth choice is subjective with no objective answer.
- **Fix:** Removed KDE entirely for discrete/bounded data

Mistake 4: Q-Q plot on bounded data

- **What we used:** Standard Q-Q plot against normal
- **Why it failed:** Theoretical quantiles go negative, data is bounded at 0. Always shows deviation at lower tail, regardless of shape.
- **Fix:** Use ECDF instead for bounded data. Q-Q only for unbounded variables where both tails have room (like temperature).

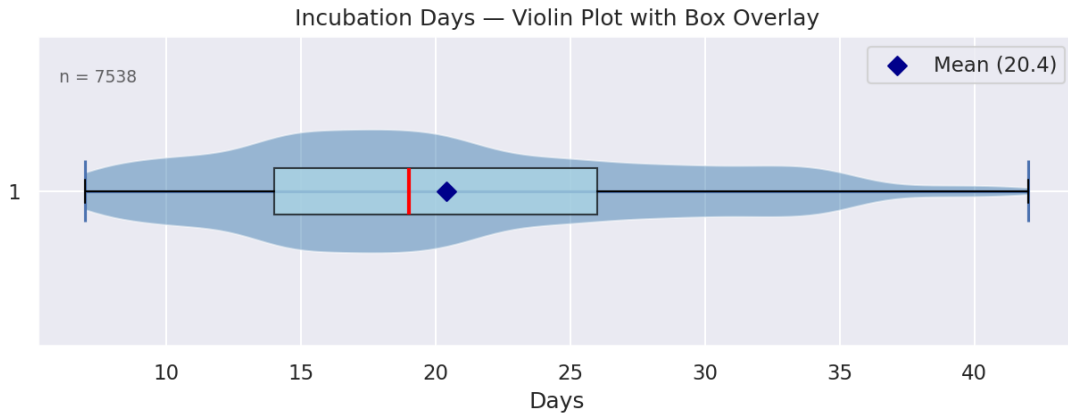


Figure 1: Incubation days — violin plot with box overlay. Mean (20.4) > Median (19).

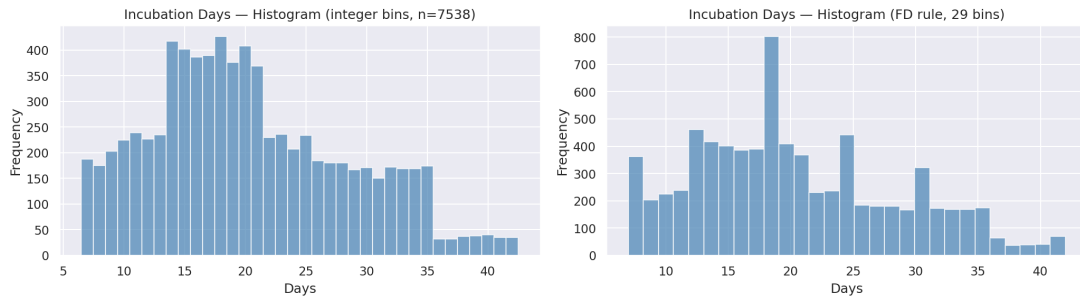


Figure 2: Incubation days — integer-aligned bins (left) and FD-rule (right).

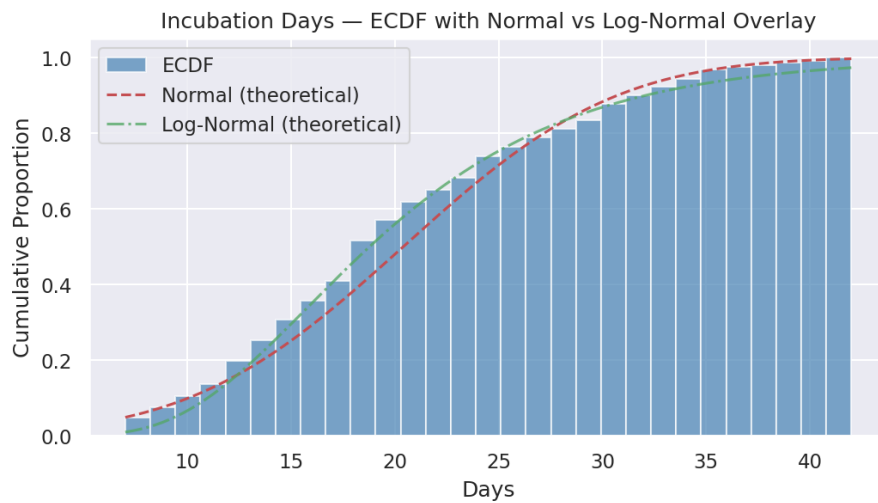


Figure 3: Incubation days ECDF — the blue line deviates from normal (red dash), but follows the log-normal (green dash-dot) more closely.

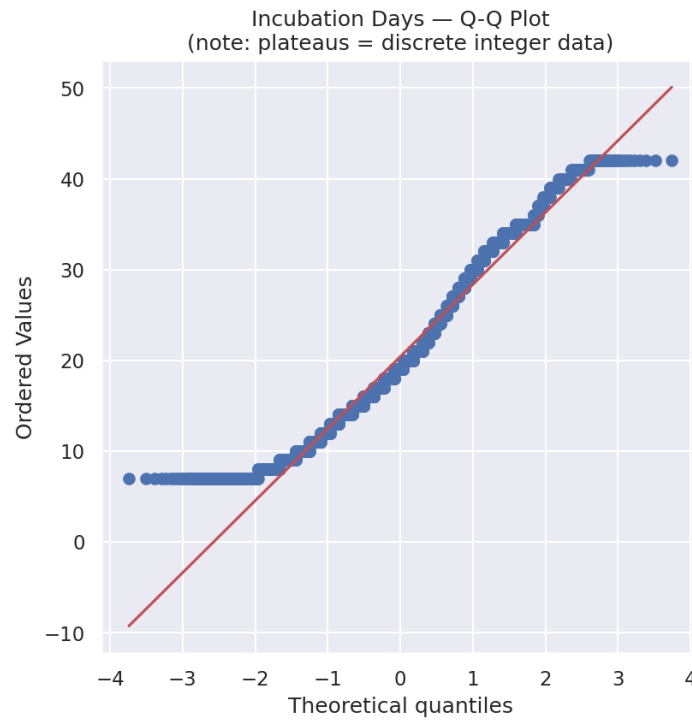


Figure 4: Incubation days Q-Q plot. Points curve away from the diagonal at extremes.

Final Results

Questionnaire result:

- User observed: right-skewed, thin tail, mean > median, not normal
- Numbers: skew=0.47, kurtosis=-0.49 (lighter tails + flatter peak than normal), mean=20.4 vs median=19
- Shapiro-Wilk: $p \approx 0.000$ — but $n=7538$ exceeds the reliable range. For $n > 5000$, any trivial deviation yields $p < 0.001$. Better: rely on $|\text{skew}| < 0.5$ and visual ECDF/Q-Q assessment.
- **Center:** Median (19d) — skew is moderate but present
- **Spread:** IQR (12d) or MAD (5d) — std (8.1d) assumes symmetry

5.2 D2: Hospital Days (D2_hospital_days.py, D2b_hospital_investigation.py)

The Bimodal Discovery

The overall histogram of hospital_days appeared bimodal with dips at days 4 and 9.

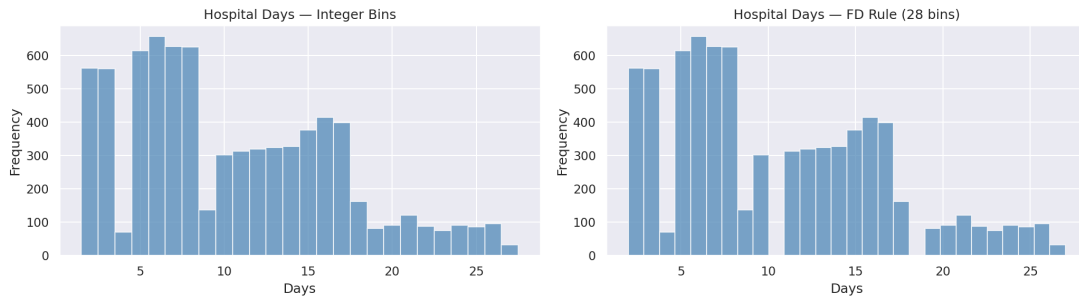


Figure 5: Hospital days histogram — appears bimodal with dips at days 4 and 9.

We investigated four possible explanations (D2b):

Conclusion: Severity explains it perfectly. Each level has a distinct, non-overlapping center. The "bimodal" shape is an **aggregation artifact** mixing four unimodal distributions.

Other factors tested and ruled out:

- Outcome: weak effect (tracks severity naturally)
- Syndrome: HPS vs HFRS nearly identical (10.3 vs 10.5 days)
- Age group: all ~ 10.4 days — no effect

Decision: Do NOT smooth the histogram. The bumps are real signal.

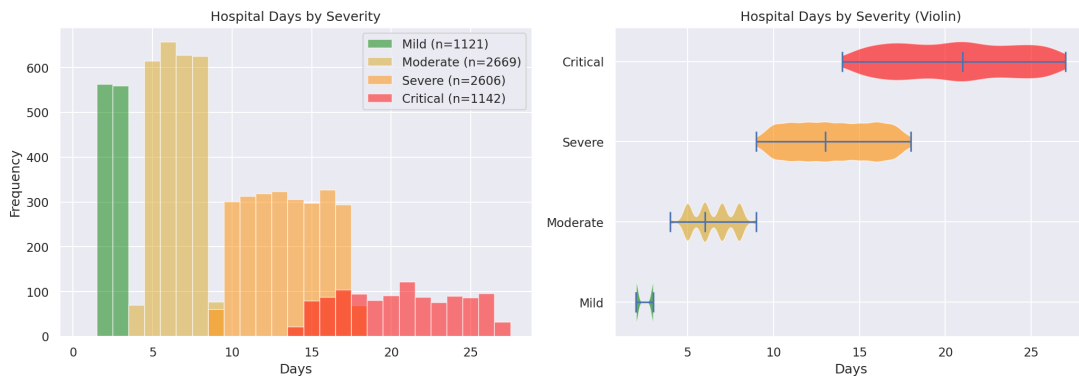


Figure 6: Hospital days by severity — each group is cleanly separated.

5.3 D3: ICU Days (D3_icu_days.py, D3b_icu_investigation.py)

Spike-and-Slab Distribution

50.3% of patients have `icu_days = 0`. The minimum stay for admitted patients is 3 days. ICU admission is perfectly predicted by severity:

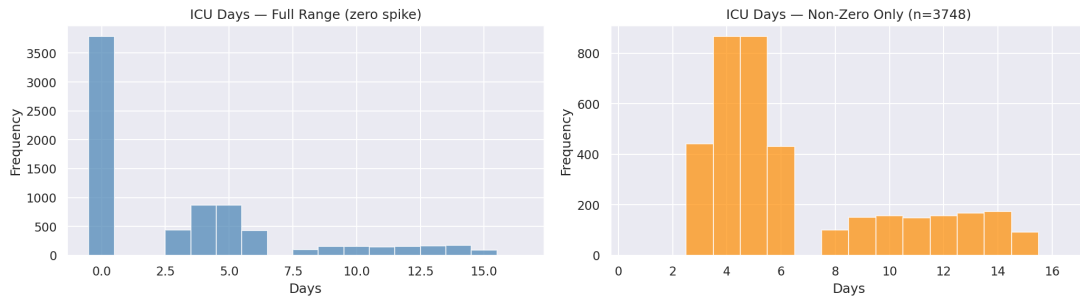


Figure 7: ICU days — massive zero spike (left). Non-zero only distribution (right).

Decision: Split into two separate questions — (1) Who goes to ICU? (binary), (2) How long do admitted patients stay? (continuous, non-zero only).

5.4 D4: Average Temperature (D4_avg_temp_c.py)

First near-normal variable. Skew=-0.15, mean $\approx$ median.

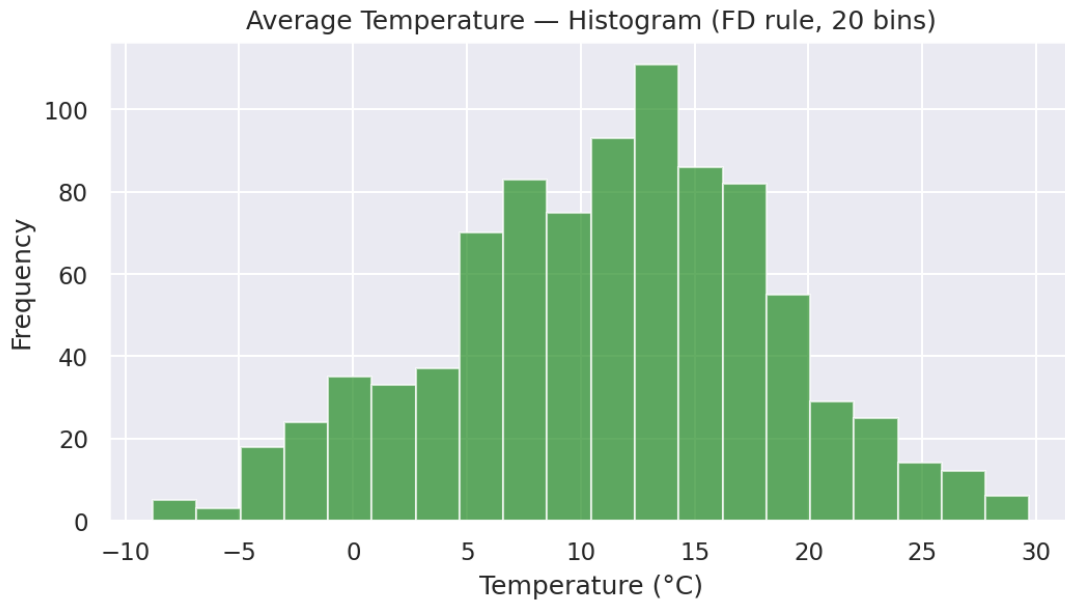


Figure 8: Temperature — approximately normal distribution.

Only outlier: -8.8°C in Ostrobothnia, Finland (real, kept).

Lesson: This is what a normal distribution looks like — useful reference for comparing with the skewed variables.

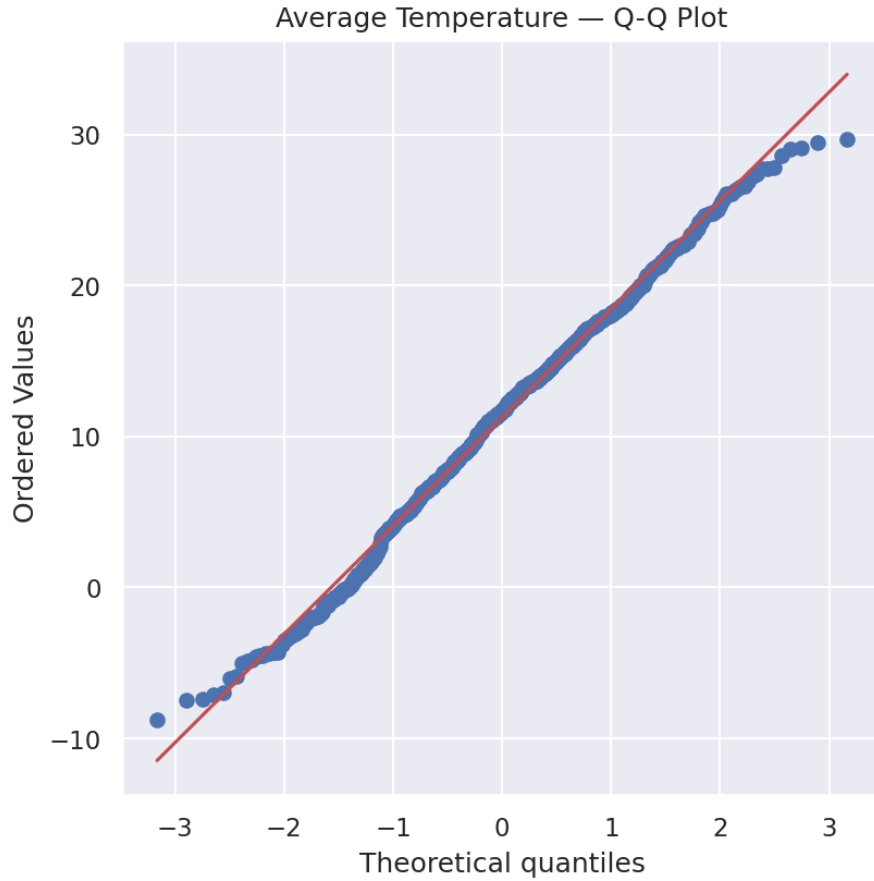


Figure 9: Temperature Q-Q plot — follows the diagonal closely (nearly normal).

5.5 D5: Rainfall (D5_rainfall_mm.py)

Right-skewed (skew=1.80), heavy-tailed (kurtosis=5.05). Mean=253mm, Median=195mm. Biome explains most variation.

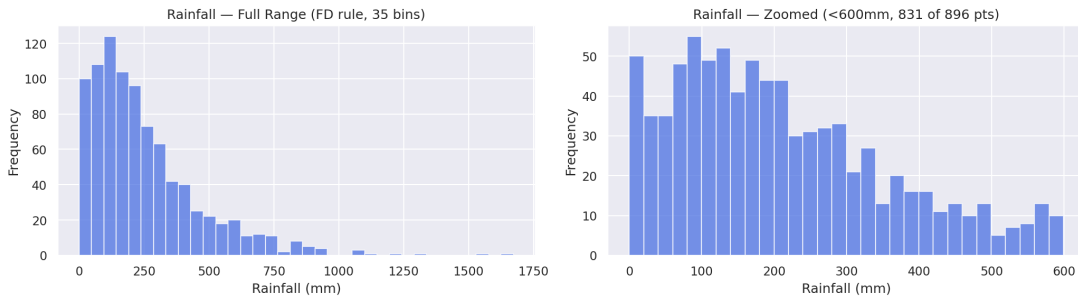


Figure 10: Rainfall — full range (left) and zoomed to <600mm (right).

5.6 D6: Rodent Abundance (D6_rodent_abundance.py)

Most symmetric variable (skew=0.02). Bounded [0,1] but nearly identical across all 14 biomes (range 0.64–0.72). Unlike rainfall (10x biome variation), rodent abundance is constant — suggesting local factors matter more than climate. **Caveat:** This assumes comparable sampling protocols (trap effort, survey area) across biomes. If methods differed, the apparent constancy could partly reflect methodological consistency.

5.7 D7: Confirmed Cases (D7_confirmed_cases.py)

Most extreme skew: mean=1793, median=58 (31x gap!). Skew=2.82, kurtosis=7.14.

Top 20 highest case years are all China HFRS (17k–24k/year since 1970s).

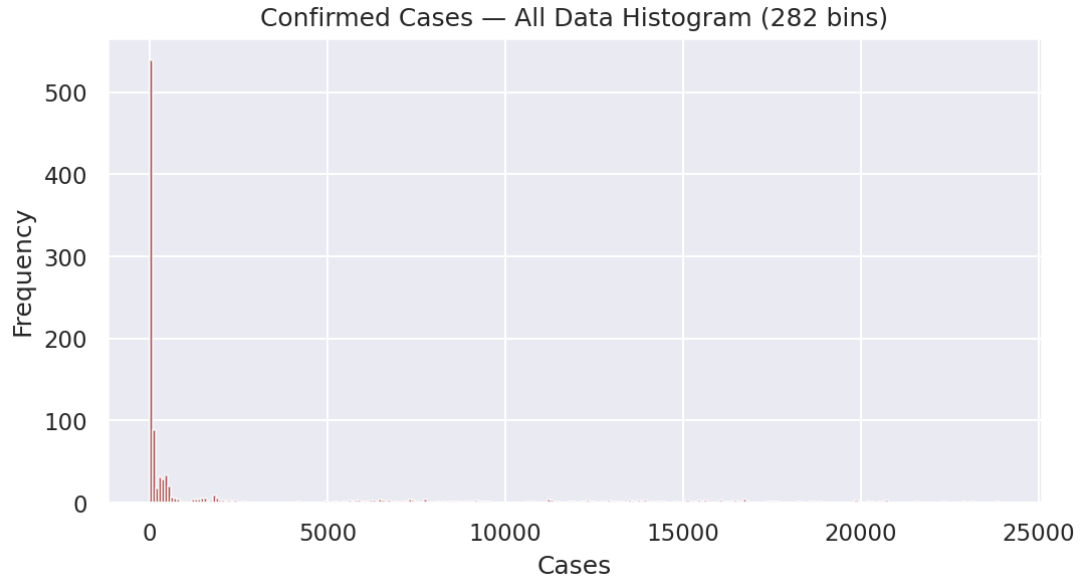


Figure 11: Confirmed cases — full range (left, massive tail) and zoomed (right, bulk of data).

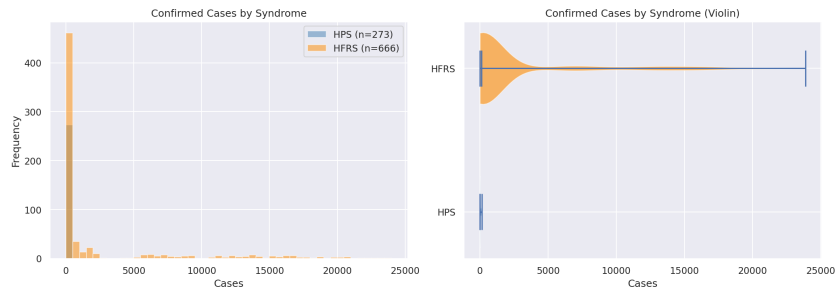


Figure 12: Confirmed cases by syndrome — HPS vs HFRS. HFRS dominates.

The Underreporting Question

Is the 100x difference between AMRO (mean=30) and WPRO (mean=5358) real? Consistent data across decades and multiple countries supports real biological differences in transmission ecology (rodent host distribution).

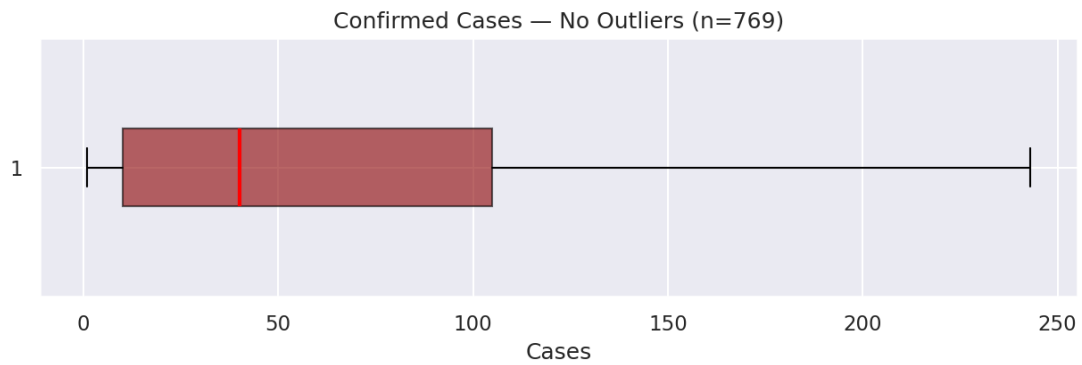


Figure 13: Confirmed cases — no-outliers version (IQR-filtered, 170 points removed).

5.8 D8: Case Fatality Rate (D8_cfr.py)

Bimodal by syndrome. HPS (mean=28.1%) is 20x more lethal than HFRS (mean=1.4%).

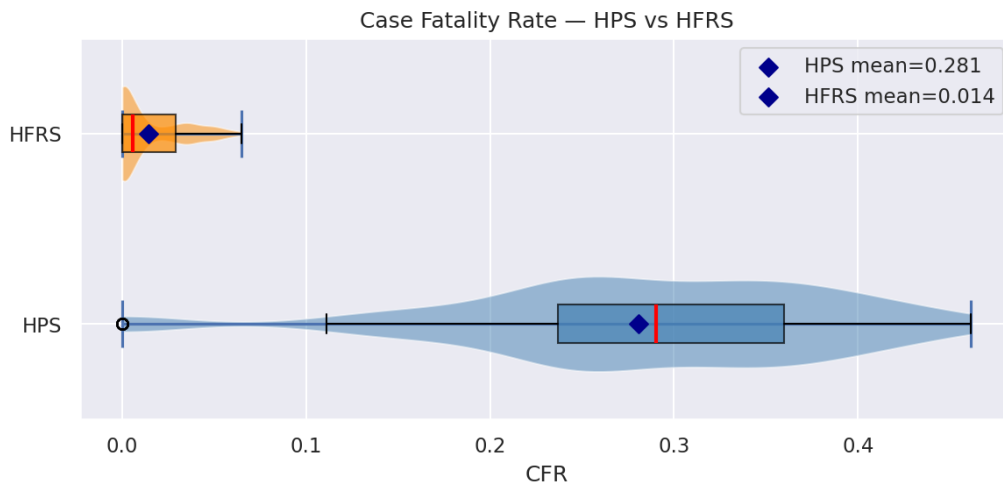


Figure 14: Case fatality rate — HPS vs HFRS side by side. The distributions barely overlap.

Key insight: The overall CFR (9.2%) represents neither group. Never report CFR as a single number for diseases with different syndromes. **Note:** CFR differences may partly reflect healthcare access or time-to-treatment, not just virology. If variables for these existed, stratifying by them would strengthen the analysis.

6 Phase E: Comparative Analysis

6.1 E1 – Comparative Violins (E1_comparative_violins.py)

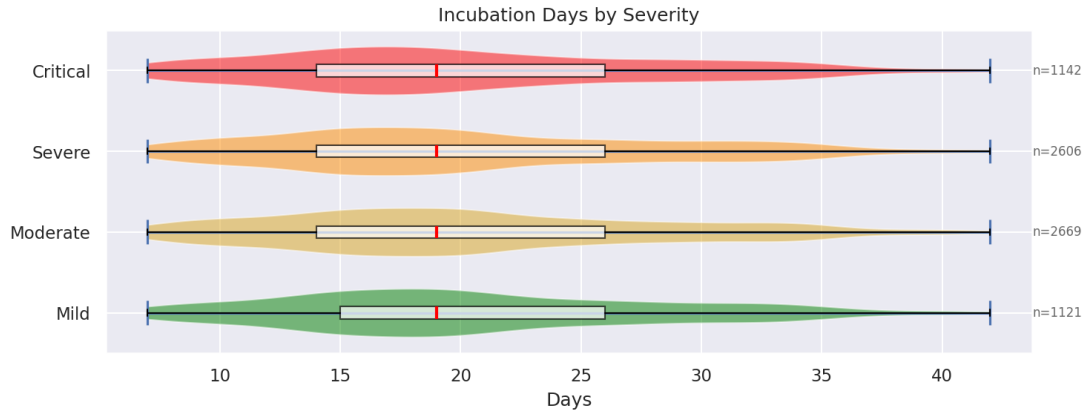


Figure 15: Incubation days by severity — surprisingly similar across all four levels.

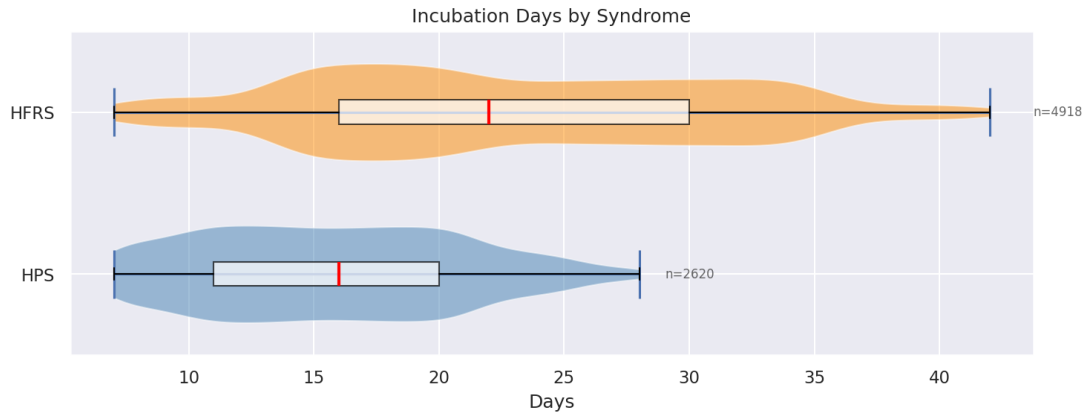


Figure 16: Incubation days by syndrome — HPS (16d) shorter than HFRS (22d).

Sample size labels ($n=...$) are shown on each violin. For any subgroup with $n < 30$, violin KDE becomes unstable and a box-only plot should be used instead.

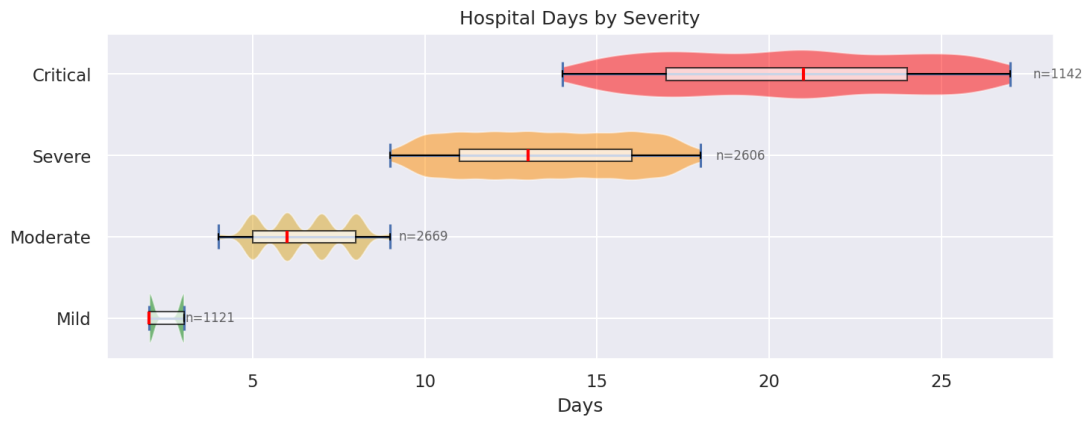


Figure 17: Hospital days by severity — clean separation.

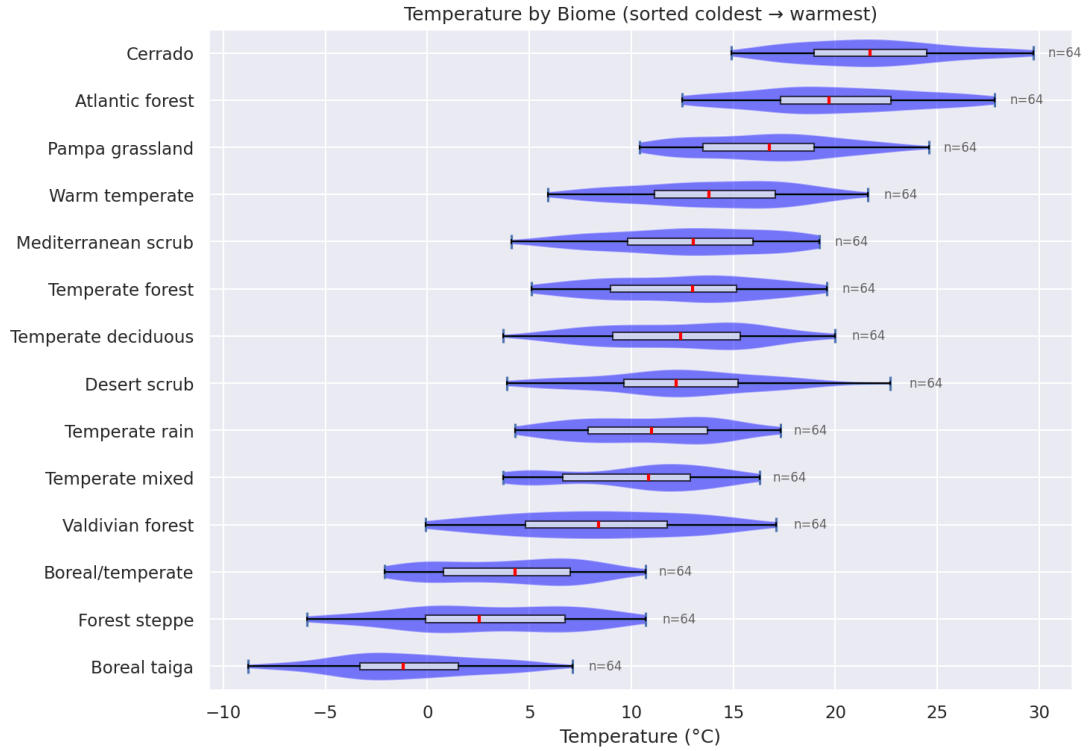


Figure 18: Temperature by biome, sorted coldest to warmest.

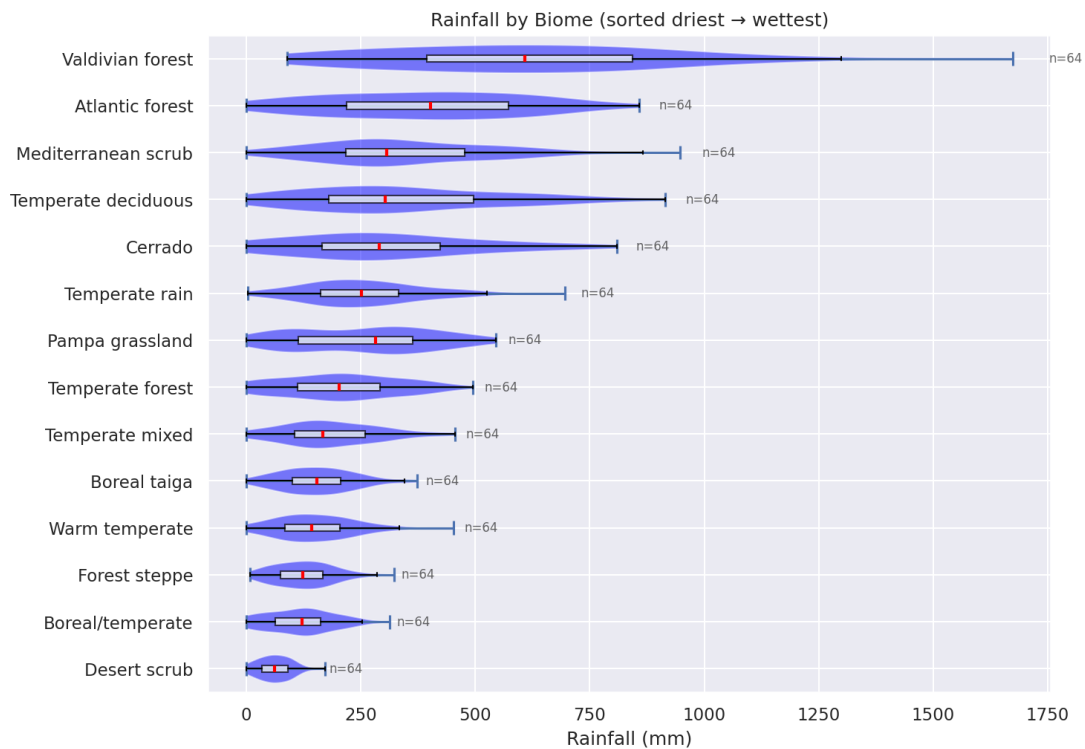


Figure 19: Rainfall by biome, sorted driest to wettest. Desert (63mm) to Valdivian forest (630mm).

6.2 E2 – Bar Chart Critique (E2_bar_chart_critique.py)

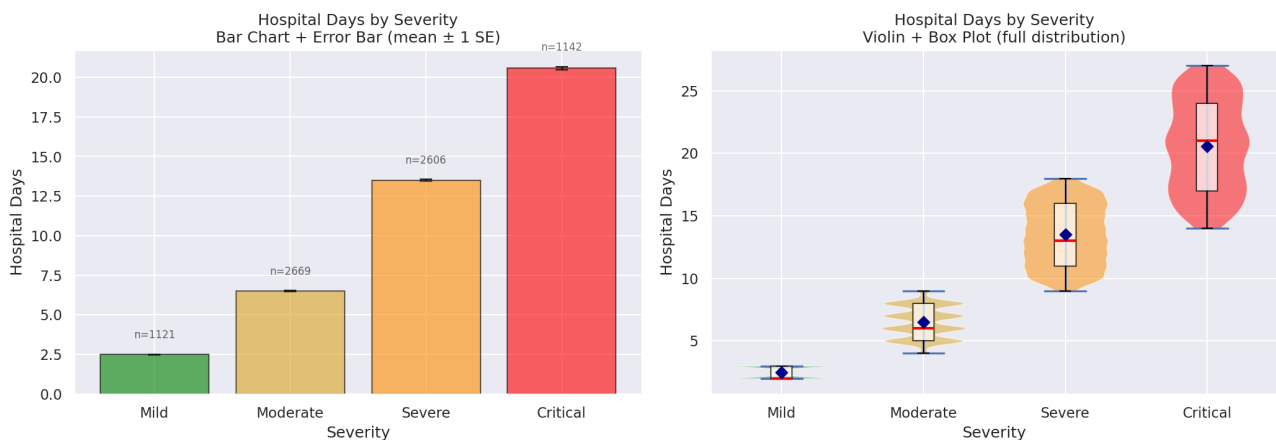


Figure 20: Side-by-side comparison: bar chart + error bars (left) vs violin + box (right).

The bar chart (left) looks clean, but hides:

1. **Sample size:** n=1121 vs n=1142 — same bar, different reliability
2. **Distribution shape:** Each group has a different shape (tight, spread, skewed)
3. **Multimodality:** If a group were bimodal, you'd never know
4. **Outliers:** Hidden completely
5. **Mean assumption:** Error bar assumes symmetric normal — hospital_days violates this

The violin + box (right) conveys: n, min, Q1, median, Q3, max, shape, skew, tails.

Bottom line: Bar charts are not "cleaner." They are less honest.

7 Master Summary Table

Decision guide: For symmetric variables (temperature, rodent abundance), mean + std is appropriate. For skewed variables (incubation, rainfall, cases), median + IQR is honest. For spike-and-slab (ICU), report separately. For bimodal (CFR), report per syndrome.

8 Key Lessons

8.1 Statistical

1. Mean vs median gap is the quickest asymmetry test
2. For skewed data: median + IQR; for symmetric: mean + std
3. Kurtosis reveals tail behavior independent of skew
4. Shapiro-Wilk is unreliable for $n > 5000$
5. Small n (outbreaks: 20) produces unreliable estimates

8.2 Visualization

1. KDE and Q-Q assume continuous data — misbehave on discrete/bounded
2. Integer data needs integer-aligned bins, not sqrt rule
3. ECDF with theoretical normal overlay is more robust than Q-Q for bounded data
4. Violin + box conveys more information than bar + error bar
5. Always check subgroups before describing shape
6. Provide both full-range and no-outliers versions for skewed data

8.3 Epidemiological

1. Incubation is similar across severity but differs by syndrome (HPS shorter)
2. Hospital/ICU stays differ by severity with clean separation
3. HPS (28.1% CFR) is 20x more lethal than HFRS (1.4% CFR)
4. China HFRS drives the extreme tail in confirmed cases
5. Rodent abundance is constant across biomes; rainfall varies 10x